

Exam I Review Topics

1. Points on a Line
 - Absolute Value
 - Solving Absolute Value Equations & Inequalities ExI-2
 - Solving Equations and Inequalities 1.1-21, 1.1-26
 - Interval Notation

2. Cartesian Coordinates
 - Points in Plane ExI-1
 - Distance Between Points ExI-1
 - Mid-Point of a Line Segment ExI-1
 - Equations of Circles 1.1-41

3. Equations of Lines
 - Slope of a Line
 - Equation of Line through a Point (x_0, y_0) with Slope m ExI-1,1.2-12
 - Parallel and Perpendicular Lines 1.2-14
 - Solving for the Intersection Set of Two Lines ExI-3,1.2-35

4. Functions
 - Domain of a Function ExI-4,1.3-6
 - Functional Notation ExI-4
 - Vertical Line Test
 - Composition of Two Functions 1.3-35
 - Graphs of Functions
 - Even/Odd Functions 1.3-29
 - Special Values of Trig Functions 1.1-44

5. Inverse of Function
 - Definition of the Inverse
 - Horizontal Line Test
 - Solving Algebraically for the Inverse Function ExI-5,1.4-15
 - Graphing the Inverse Function ExI-5,1.4-34
 - Functional Relationships between a Function and its Inverse
$$f^{-1}(f(x)) = x, \quad f(f^{-1}(x)) = x$$

6. Inverse Trig Functions
 - Domain Restrictions for Sine, Cosine, Tangent
 - Graph of Inverse Sine, Inverse Cosine, Inverse Tangent
 - Functional Relationships ExI-6,ExI-7,1.4-25,1.4-41

7.	Limit of a Function	
	Estimating Limit	
	Tabular Approach	2.1-21,2.1-35
	Graphical Approach	ExI-8,2.1-(1-6)
	One-Side Limit	ExI-8,2.1-(1-6)
8.	Algebraic Computation of Limits	
	Basic Properties and Rules	
	Limits of Polynomials	ExI-9,2.2-(1-3)
	Limits of Rational Expressions via Algebraic Reduction	ExI-9,2.2-(11-14)
	Limits of Trig Functions	2.2-(7-9)
	Special Limits $\lim_{x \rightarrow 0} \frac{\sin x}{x}$, $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x}$	ExI-9,2.2-(21-25)
	Sandwich/Squeeze Rule	
9.	Continuity	
	Definition of Continuity at a Point	2.3-(15-16)
	Properties of Continuous Functions	
	One-Side Continuity at a Point	
	Continuity on an Interval	2.3-(23-24)
	Max/Min Theorem	
	Intermediate Value Theorem	2.3-(27-28)
10.	Exponential Functions	
	Properties	
	Algebraic Properties	
	Solving Exponential Equations	2.4-(31-33)
11.	Logarithm Functions == Inverses of Exponential Functions	
	Algebraic Properties	
	Properties	
	Graph	
	Domain, Range, Asymptote, Monotonicity	
	Solving Logarithmic Equations	2.4-(39-40)
	Natural Logarithm	
	Properties	
12.	Exponential Growth/Decay	2.4-66